MATH 127 Calculus for the Sciences

Lecture 22

Today's lecture

Last time
Limit at infinity
Indeterminant form

This time
Course note coverage Section 3.2.2 and 3.2.3
More on indeterminate forms.

Quiz 6 is today

Coverage

1. Everything covered up to the midterm
2. Continuity: for which inputs is a function is continuous? (check using definition or continuity rule)
3. Evaluating limits: using graph, or using algebra
4. Determinate form: identify them, evaluate them
5. Indeterminate form: identify them (you don't need to know how to evaluate them)

Recap

The following are determinate forms we have seen so far, and two indeterminate forms that we learn how to deal with today.

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
$\frac{1}{0^-}, \frac{\infty}{0^-}$	$-\infty$
$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0
Indeterminate form	**tricks**
$\frac{\infty}{\infty}, \frac{0}{0}$	will see today
$0 \cdot \infty$	will see on Friday
$0^0, \infty^0, 1^\infty$	will see on Friday

Simplify indeterminate form

Example Consider

$$\lim_{x\to 0} \frac{(x-2)^2-4}{3x}$$

Which indeterminate form is this? Compute by simplifying it.

"$\frac{(0-2)^2-4}{3\cdot 0} = \frac{4-4}{0} = \frac{0}{0}$" This limit has indeterminate form of $\frac{0}{0}$.

$$\lim_{x\to 0} \frac{(x-2)^2-4}{3x} = \lim_{x\to 0} \frac{(x^2-4x+4)-4}{3x} = \lim_{x\to 0} \frac{x^2-4x}{3x}$$

$$= \lim_{x\to 0} \frac{x-4}{3} = \frac{\boxed{\lim_{x\to 0} x} - \boxed{\lim_{x\to 0} 4}}{\boxed{\lim_{x\to 0} 3}}$$

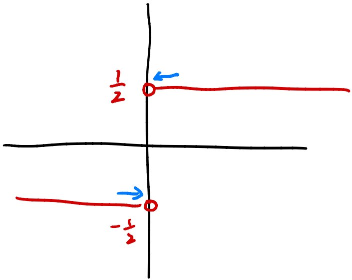
$$= \frac{0-4}{3} = -\frac{4}{3}$$

Simplify indeterminate form

Example Evaluate the limit

$$\lim_{x\to 0} \frac{|x|}{2x}$$

1. Is 0 in the domain of this function? No
2. How do we simplify this function?



By the graph,

$$\lim_{x\to 0^-} \frac{|x|}{2x} = -\frac{1}{2}, \quad \lim_{x\to 0^+} \frac{|x|}{2x} = \frac{1}{2}$$

Hence the original limit does not exist.

$$|x| = \begin{cases} \boxed{x \quad \text{if } x \geq 0} \\ \boxed{-x \quad \text{if } x < 0}. \end{cases}$$

$$\lim_{x\to 0^-} \frac{|x|}{2x} = \lim_{x\to 0^-} \frac{-x}{2x} = \lim_{x\to 0^-} \frac{-1}{2} = -\frac{1}{2}$$

$$\lim_{x\to 0^+} \frac{|x|}{2x} = \lim_{x\to 0^+} \frac{x}{2x} = \lim_{x\to 0^+} \frac{1}{2} = \frac{1}{2}$$

Multiply by a conjugate

Recall the formula

$$(X+Y)(X-Y) = X^2 - Y^2.$$

This means

$$(\sqrt{A}+\sqrt{B})(\sqrt{A}-\sqrt{B}) = \boxed{\sqrt{A}^2 - \sqrt{B}^2 = A - B}.$$

Trick If you start with an indeterminate form with something in the denominator/numerator like

$$(\sqrt{A}+\sqrt{B})$$

you can multiply both denominator and numerator by its **conjugate**

$$(\sqrt{A}-\sqrt{B})$$

and things might magically work out.

Example: Multiply by a conjugate

Example Evaluate

$$\lim_{x\to 1^+} \frac{(x-1)^2}{\sqrt{x+2}-\sqrt{2x+1}}$$

Try multiplying the conjugate on both the denominator and numerator.

This limit has indeterminate form $\frac{0}{0}$

$$\lim_{x\to 1^+} \frac{(x-1)^2}{\sqrt{x+2}-\sqrt{2x+1}} = \lim_{x\to 1^+} \frac{(x-1)^2\,(\sqrt{x+2}+\sqrt{2x+1})}{(\sqrt{x+2}-\sqrt{2x+1})(\sqrt{x+2}+\sqrt{2x+1})}$$

$$= \lim_{x\to 1^+} \frac{(x-1)^2\,\boxed{(\sqrt{x+2}+\sqrt{2x+1})}}{(x+2)-(2x+1)}$$

product rule

$$= 2\cdot\sqrt{3}\,\lim_{x\to 1^+} \frac{(x-1)^2}{-x+1}$$

$$= 2\sqrt{3}\,\lim_{x\to 1^+} \frac{(x-1)^2}{-(x-1)}$$

$$= 2\sqrt{3}\,\lim_{x\to 1^+} -(x-1) = 2\sqrt{3}\cdot 0 = 0$$

L'Hospital's rule

Theorem (L'Hospital's rule, or L'Hôpital's rule if you are fancy)
If

$$\lim_{x\to a}\frac{f(x)}{g(x)}$$

has indeterminate form $\boxed{\frac{0}{0} \text{ or } \frac{\infty}{\infty}}$, *then*

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

as long as $\boxed{\text{RHS exists}}$ *or equal to* ∞ *or* $-\infty$.

Example: L'Hospital's rule

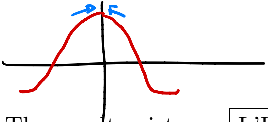
Example Use L'Hospital's rule to solve

$$\lim_{x \to 0} \frac{\sin x}{2x}$$

1. The limit is of indeterminate form $\boxed{\frac{0}{0}}$
2. Taking derivative of both denominator and numerator, we get

$$\lim_{x \to 0} \boxed{\frac{\cos x}{2}}$$

3. This new limit is no longer a indeterminate form, and evaluates to



$$\lim_{x \to 0} \boxed{\frac{\cos x}{2}} = \boxed{\frac{1}{2}}$$

4. The result exists, so $\boxed{\text{L'Hospital's rule}}$ applies, and we conclude the original limit is $\boxed{\frac{1}{2}}$.

Example: L'Hospital's rule

Example Use L'Hospital's rule to solve

$$\lim_{x\to\infty} \boxed{\frac{\sqrt{x^2-1}}{x}}$$

The limit is of indeterminate form $\boxed{\frac{\infty}{\infty}}$

$$\lim_{x\to\infty} \frac{(\sqrt{x^2-1})'}{(x)'} = \lim_{x\to\infty} \frac{(x^2-1)'}{2\sqrt{x^2-1}} \Big/ 1 = \lim_{x\to\infty} \frac{\cancel{2}x}{\cancel{2}\sqrt{x^2-1}}$$

The limit is still of form $\boxed{\frac{\infty}{\infty}}$

$$\lim_{x\to\infty} \frac{(x)'}{(\sqrt{x^2-1})'} = \lim_{x\to\infty} 1 \Big/ \frac{(x^2-1)'}{2\sqrt{x^2-1}} = \lim_{x\to\infty} \boxed{\frac{\sqrt{x^2-1}}{x}}$$

L'Hospital's rule does not work.

Example: L'Hospital's rule not working

Example We saw L'Hospital's rule doesn't work for

$$\lim_{x\to\infty} \frac{\sqrt{x^2-1}}{x}$$

What else can we try to find the limit?

1. Simplification?

~~2.~~ Multiply by conjugates?

$$\text{original limit} = \lim_{x\to\infty} \sqrt{\frac{x^2-1}{x^2}} = \sqrt{\lim_{x\to\infty} \frac{x^2-1}{x^2}}$$

$$= \sqrt{\lim_{x\to\infty} 1 - \frac{1}{x^2}}$$

$$= \sqrt{1} = 1.$$

example:L'Hospital's rule

Example Use L'Hospital's rule to solve

$$\lim_{x\to\infty} \frac{x+\sin x}{x}$$

This is of indeterminate form $\frac{\infty}{\infty}$

$$\lim_{x\to\infty} \frac{(x+\sin x)'}{(x)'} = \lim_{x\to\infty} \frac{1+\cos x}{1}$$

$$= \lim_{x\to\infty} 1+\cos x .$$

The limit does not exist.

Example: L'Hospital's rule not working

Example We saw L'Hospital's rule doesn't work for

$$\lim_{x\to\infty} \frac{x+\sin x}{x}$$

What else can we try to find the limit?

$$\lim_{x\to\infty} \frac{x+\sin x}{x} = \lim_{x\to\infty}\left(\frac{x}{x} + \frac{\sin x}{x}\right)$$

$$= 1 + \lim_{x\to\infty} \frac{\sin x}{x}$$

$$= 1 + 0$$

because $\sin x$ is bonded by $[-1, 1]$. so as $x\to\infty$, $\frac{\sin x}{x} \to 0$.

$$= 1$$

Summary

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
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$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0

Indeterminate form	tricks
$\frac{\infty}{\infty}, \frac{0}{0}$	simplify multiply conjugate L'Hospital's rule
$0 \cdot \infty$	will see on Friday
$0^0, \infty^0, 1^\infty$	will see on Friday